**STAT 709 Mathematical Statistics** Lecture 4 - Source draft

# Lecture 4: Basics of Probability Theory 1

*Lecturer: Yiqiao Zhong*

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*These notes are used for STAT 709 "Mathematical Statistics" at the University of Wisconsin–Madison. They are a learning aid rather than a substitute for a textbook; corrections and questions are welcome.*

**High priority** *From integration to probability*

Lectures 2 and 3 supplied random variables, their laws, and integration. We now give the probability integral its usual name: **expectation**. The central question is how information about a distribution becomes a useful numerical summary or a probability bound. An average loss, a variance, and an expected number of triangles in a graph are all expectations, even though their formulas may look different.

We first learn to compute expectations from laws and tails. We then study norms and inequalities, and use them in examples involving random graphs, information theory, and collinearity. Independence and conditional expectation are developed further in Lecture 5; in this lecture, any independence used in a specific model is stated explicitly.

**Recap from earlier lectures**

*From Lectures 1–3: integrals and inequalities*

Reminder only: this block introduces no new exam requirements.

- **Integrability**. $\int |f|\, d\nu < \infty$ makes the signed integral finite. Used in Section 1.
- **Convergence theorems**. MCT uses nonnegative monotonicity; DCT uses an integrable bound; Fatou gives an inequality. Used in Section 2.
- **Tonelli**. Nonnegative integrands allow the order of integration to be exchanged. Used in the tail identity.
- **Radon–Nikodym derivative**. A density depends on its reference measure, which can include atoms. Used in Section 1.
- **Convexity and norm inequalities**. Convexity compares averages; Cauchy–Schwarz bounds inner products by norms. Used in Section 2.

## Learning goals

- Distinguish an integrable random variable from one with an infinite or undefined expectation, and compute an expectation using its law.
- Recognize discrete, absolutely continuous, mixed, and singular continuous laws without treating every continuous CDF as a density.
- Apply Tonelli, Hölder, Cauchy–Schwarz, Jensen, Markov, and dominated convergence with their required hypotheses.
- Check boundary assumptions, equality cases, and covariance normalizations in statistical applications.

**Priority guide**

**High** Core concepts to master for this lecture; will be directly examined.

**Mid** Supporting concepts useful in other lectures; in scope of examinations.

**Low** Enrichment supporting understanding; will not be examined.

**Further reading** Optional self-study, references, extensions, and historical notes.

# 1 Expectation

**High priority** *Definition and the role of the probability measure*

**Definition 1** (Expectation and integrability)**.** *Let $X : \Omega \to \mathbb{R}$ be a random variable on $(\Omega, \mathcal{F}, P)$. If $X \geq 0$ almost surely, define $\mathbb{E}\, X = \int_\Omega X\, dP \in [0, \infty]$. For a signed $X$, write $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$. Then $\mathbb{E}\, X = \mathbb{E}\, X^+ - \mathbb{E}\, X^-$ is defined if at least one term is finite; $\infty - \infty$ is undefined. We call $X$* ***integrable****, or say it has finite expectation, when*

$$\mathbb{E}\,|X| = \int_\Omega |X(\omega)|\, dP(\omega) < \infty.$$

*In that case both terms are finite and $\mathbb{E}\, X = \int_\Omega X\, dP \in \mathbb{R}$.*

Linearity holds for integrable random variables and does not require independence. Expectation also preserves inequalities: if $X \leq Y$ a.s. and both are integrable, then $Y - X \geq 0$, so positivity of the integral gives $\mathbb{E}\, Y - \mathbb{E}\, X = \mathbb{E}(Y - X) \geq 0$. For nonnegative variables the corresponding monotonicity also holds with infinite expectations.

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**Instructor's Note** *A unified definition*

In undergraduate probability, expectation is a probability-weighted sum of outcomes for discrete random variables and a density-weighted integral for continuous random variables. Measure theory provides a unified definition. The weighting is "absorbed" into an abstract probability measure $P$ without specifying the base measure.

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## 1.1 Changing from outcomes to the law of a random variable

**High priority** *Two domains, the same expectation*

**Definition 2** (Distribution or law)**.** *For a real-valued $X$, its* ***law*** *is the probability measure $\mu_X = P \circ X^{-1}$ on $(\mathbb{R}, \mathcal{B})$ given by*

$$\mu_X(B) = P\{\omega : X(\omega) \in B\}, \qquad B \in \mathcal{B}.$$

*The* ***cumulative distribution function (CDF)*** *is $F_X(t) = P(X \leq t) = \mu_X((-\infty, t])$.*

For a Borel function $g : \mathbb{R} \to \mathbb{R}$, the change-of-variables identity is

$$\mathbb{E}[g(X)] = \int_\Omega g(X(\omega))\, dP(\omega) = \int_{\mathbb{R}} g(x)\, d\mu_X(x). \tag{1}$$

It holds for nonnegative $g$, allowing $+\infty$, and for integrable $g(X)$; more generally it holds whenever the positive/negative-part definition is unambiguous. In particular, $\mathbb{E}\,|g(X)| = \int |g|\, d\mu_X$. The measure on the right lives on possible values of $X$; it is not the original measure $P$ on outcomes. This identity does not require a density or a one-to-one map. A vector-valued $X$ uses the same argument with $\mathbb{R}^d$ in place of $\mathbb{R}$.

**Exercise prompt: begin with simple functions.** Let $g = \sum_{j=1}^{m} c_j \mathbf{1}_{B_j}$, where $c_j \geq 0$ and the $B_j$ are disjoint Borel sets. Verify (1) directly from the definition of $\mu_X$. See Exercise 1 in Section 5 for the full statement, extension to nonnegative functions, and complete solution.

The law and its CDF determine each other. For $a < b$,

$$F_X(b) - F_X(a) = P(a < X \leq b) = \int_\Omega \mathbf{1}_{(a,b]}(X(\omega))\, dP(\omega) = \mu_X((a,b]). \tag{2}$$

Thus the CDF specifies the probabilities of half-open intervals; the uniqueness of the Borel probability measure with these interval values is the CDF correspondence from Lecture 2. Notice the integration domain $\Omega$ in the middle expression, rather than the interval $(a, b]$.

**Mid priority** *A mixed law retains both kinds of mass*

**Example 1** (Discrete and absolutely continuous components)**.** *Let $p_j \geq 0$, $\sum_{j\geq 0} p_j = 1$, and $\mu_0 = \sum_{j\geq 0} p_j \delta_j$, where $\delta_j$ puts unit mass at $j$. Let $h \geq 0$ be a Lebesgue density with $\int_{\mathbb{R}} h(x)\, dx = 1$. Then*

$$\mu(B) = \tfrac{1}{2}\mu_0(B) + \tfrac{1}{2}\int_B h(x)\, dx$$

*is a probability law. Its CDF and, for integrable $g$, its expectation are*

$$F(t) = \tfrac{1}{2}\sum_{j\leq t} p_j + \tfrac{1}{2}\int_{-\infty}^{t} h(x)\, dx, \qquad \int g\, d\mu = \tfrac{1}{2}\sum_{j\geq 0} p_j g(j) + \tfrac{1}{2}\int_{\mathbb{R}} g(x)h(x)\, dx.$$

*The CDF jumps by $p_j/2$ precisely at the integers with $p_j > 0$ and is continuous elsewhere. This law has neither a Lebesgue density nor a pmf that accounts for all its mass. It does have a density relative to the combined reference measure $\sum_{j\geq 0} \delta_j + dx$, as in Lecture 3.*

For a concrete picture, take $\mu = \frac{1}{2}\delta_0 + \frac{1}{2}\mathrm{Unif}([0,2])$. Then

$$F(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{2} + \frac{1}{4}t, & 0 \leq t \leq 2, \\ 1, & t > 2. \end{cases}$$

The jump at zero records an atom of mass $1/2$; the rise from $1/2$ to 1 over $(0, 2)$ records the continuous half of the mass. Integrating only the ordinary density $\frac{1}{4}\mathbf{1}_{(0,2)}$ would miss the atom.

## 1.2 A three-part decomposition of a law

**Low priority** *Optional refinement of the mixed-law picture*

Every Borel probability measure on $\mathbb{R}$ has a unique decomposition

$$\mu = \mu_{\mathrm{dis}} + \mu_{\mathrm{cont}} + \mu_{\mathrm{sing}}. \tag{3}$$

Here the three terms are finite nonnegative measures whose total masses sum to one; individually they need not be probability measures.

- The **discrete (atomic) part** is $\mu_{\mathrm{dis}} = \sum_{a\in D} \mu(\{a\})\delta_a$, where $D = \{a : \mu(\{a\}) > 0\}$ is at most countable. Its atoms give exactly the CDF jumps, since $F(a) - F(a-) = \mu(\{a\})$.
- The **absolutely continuous part** satisfies $\mu_{\mathrm{cont}}(B) = \int_B f(x)\, dx$ for some $f \geq 0$. Its total mass $\int f$ can be smaller than one. We use the legacy subscript "cont" for this part, but it means absolutely continuous with respect to Lebesgue measure, not merely a continuous CDF.

- The **singular continuous part** $\mu_{\text{sing}}$ is concentrated on a Lebesgue-null Borel set and has no atoms: $\mu_{\text{sing}}(\{a\}) = 0$ for every $a$. This distinguishes it from the entire singular part relative to Lebesgue measure, which also includes the atoms. The Cantor distribution is a standard example.[1]

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**Instructor's Note** *Three pictures of probability mass*

Think of three ways to arrange probability mass: piles at individual points, mass spread out with a density, and atomless mass on a very thin set. The piles make jumps in the CDF. For the third picture, imagine successively concentrating the mass onto thinner collections of intervals: in the limit, all the probability sits on a zero-measure set, yet no single point carries a pile. The CDF is still continuous.

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The construction and a precise proof reference are given in Section 4. No part of the high/mid calculations below requires this optional decomposition theorem.

## 1.3 Expectations from tails and Gaussian transformations

**High priority** *Tail integrals*

If $f : [0, \infty) \to [0, \infty)$ is nondecreasing and continuously differentiable with $f(0) = 0$, then

$$\mathbb{E}[f(|X|)] = \int_0^\infty f'(t) P(|X| > t)\, dt. \tag{4}$$

Both sides may be infinite. The appropriate interchange theorem is **Tonelli's theorem**, because the integrand is nonnegative; one need not assume the desired expectation is finite beforehand. In particular,

$$\mathbb{E}\,|X| = \int_0^\infty P(|X| > t)\, dt.$$

This relates the average magnitude to the entire tail curve, rather than to a single tail probability.

**Exercise prompt: justify the tail identity.** Show that $\mathbb{E}\,|X| = \int_0^\infty P(|X| > t)\, dt$ using Tonelli, including the case of infinite expectation. See Exercise 2 in Section 5 and its complete solution for the derivation and the extension to (4).

A second discrete example is drawn in the complete tail-identity solution.

**Mid priority** *Gaussian squares and their distributions*

If $Z \sim N(0, 1)$, then $Z^2$ has the **chi-squared distribution** with one degree of freedom, denoted $\chi^2_1$. If $Z_1, Z_2$ are independent $N(0, 1)$ variables, then

$$Z_1^2 + Z_2^2 \sim \chi^2_2 = \text{Exp(rate } 1/2).$$

Our convention is that $Y \sim \text{Exp(rate } \lambda)$ has density $\lambda e^{-\lambda y} \mathbf{1}_{\{y>0\}}$ and survival probability $P(Y > y) = e^{-\lambda y}$ for $y \geq 0$. Thus the squared two-dimensional standard-normal radius has rate $1/2$, not rate 1. The full density and polar-coordinate calculations are in Exercise 6 and its solution; see also the expectation examples in [3].

[1] The Cantor distribution is a probability law concentrated on the Cantor set, a subset of $[0, 1]$ of Lebesgue measure zero, yet each individual point has probability zero. Its CDF is continuous but is not the integral of a Lebesgue density. It illustrates the third term in the decomposition; its full construction is optional here.

# 2 Functional perspective and inequalities

## 2.1 Norms, moments, and covariance

**High priority** *Measure the size of a random variable*

For $1 \le p < \infty$, let $L^p(P)$ consist of random variables with $\mathbb{E}\,|X|^p < \infty$. For $p = \infty$, use the essentially bounded random variables. Their $L^p$ **norms** are

$$\|X\|_p = \begin{cases} (\mathbb{E}\,|X|^p)^{1/p}, & 1 \le p < \infty, \\ \inf\{M \ge 0 : P(|X| > M) = 0\}, & p = \infty. \end{cases}$$

An empty set in the last infimum gives $+\infty$, which means $X \notin L^\infty$. The **essential supremum** ignores exceptional values on a zero-probability set. To obtain a norm, we identify variables equal almost surely, rather than treating different values on a null set as different elements.[2]

**Theorem 1** (Hölder and Cauchy–Schwarz)**.** *Let $1 \le p, q \le \infty$ with $1/p + 1/q = 1$, interpreting $1/\infty = 0$. For $X \in L^p(P)$ and $Y \in L^q(P)$,*

$$\mathbb{E}\,|XY| \le \|X\|_p \|Y\|_q.$$

*In particular, $X, Y \in L^2(P)$ imply $\mathbb{E}\,|XY| \le (\mathbb{E}\,X^2)^{1/2}(\mathbb{E}\,Y^2)^{1/2}$.*

The proof of Hölder, and the triangle inequality (Minkowski) needed for the norm assertion, are given in Exercise 7 and its complete solution. Positivity, homogeneity, and the treatment of null sets are included there. Finite-dimensional norm intuition from Lecture 1 helps, but is not itself a proof for an arbitrary probability space.

**Example 2** (Higher moments control lower moments)**.** ***Exercise prompt.*** *Suppose $1 \le p < q < \infty$ and $\mathbb{E}\,|X|^q < \infty$. Show that $\|X\|_p \le \|X\|_q$, and hence $\mathbb{E}\,|X|^p < \infty$. See Exercise 3 in Section 5 and its complete solution.*

**Example 3** (Covariance is bounded by marginal variation)**.** *For $X, Y \in L^2(P)$, define*

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}\,X)(Y - \mathbb{E}\,Y)], \qquad \mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}\,X)^2].$$

***Exercise prompt.*** *Prove $|\mathrm{Cov}(X, Y)| \le \sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$. What happens if one variance is zero? See Exercise 4 in Section 5 and its complete solution.*

## 2.2 Convexity and probability bounds

**High priority** *Jensen's inequality*

**Theorem 2** (Jensen)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a finite convex function and let $X$ be integrable. Then $\mathbb{E}\,\varphi(X)$ is well defined in $(-\infty, \infty]$, and*

$$\varphi(\mathbb{E}\,X) \le \mathbb{E}\,\varphi(X).$$

*If $\varphi$ is strictly convex, equality holds if and only if $X = \mathbb{E}\,X$ almost surely.*

[2] An equivalence class here is the collection of all random variables equal to one another with probability one. The zero element consists of variables that are zero almost surely, even if a chosen representative is nonzero on a null set. This is why norm zero means equality almost surely.

*Proof.* Put $m = \mathbb{E}\,X$. A finite convex function on $\mathbb{R}$ has a supporting line at $m$: for some finite $a$, $\varphi(x) \geq \varphi(m) + a(x-m)$ for all $x$. Indeed, every left secant slope at $m$ is at most every right secant slope, so choose $a$ between their supremum and infimum. These bounds are finite by comparison with secants through $m-1$ and $m+1$. The affine lower bound is integrable, so the negative part of $\varphi(X)$ has finite expectation. Taking expectations proves the inequality even if the positive part has infinite expectation.

For strict convexity, the supporting line touches the graph only at $m$; a second contact would force equality along the joining segment, contrary to strict convexity. Therefore the nonnegative gap $\varphi(X)-\varphi(m)-a(X-m)$ vanishes almost surely precisely when $X = m$ almost surely. A nonnegative random variable has expectation zero only if it vanishes almost surely: apply positivity to $\mathbb{E}\,V \geq k^{-1}P(V \geq k^{-1})$ for every positive integer $k$ and take the countable union. This proves the equality assertion; the converse is immediate. □

The geometric question is whether averaging before applying a convex function can give a larger value than applying it first and then averaging. For $P(X=-1) = P(X=1) = 1/2$ and $\varphi(x) = x^2$,

$$\varphi(\mathbb{E}\,X) = 0, \qquad \mathbb{E}\,\varphi(X) = 1.$$

More generally, for $P(X=x) = 1-s$ and $P(X=y) = s$,

$$(1-s)x^2 + sy^2 - \big((1-s)x + sy\big)^2 = s(1-s)(x-y)^2 \geq 0.$$

The graph lies below the chord: the gap is zero for a degenerate mixture ($s = 0$ or $1$) or identical endpoints, and positive otherwise. This calculation illustrates Jensen for a two-point law; the theorem above covers arbitrary integrable random variables.

**High priority** *Markov's inequality*

**Theorem 3** (Markov)**.** *For a nonnegative random variable $X$ and $a > 0$,*

$$P(X \geq a) \leq \frac{\mathbb{E}\,X}{a}.$$

*The bound is useful when $\mathbb{E}\,X < \infty$; with $\mathbb{E}\,X = \infty$ it remains valid but gives no information.*

*Proof.* Pointwise, $a\mathbf{1}_{\{X \geq a\}} \leq X\mathbf{1}_{\{X \geq a\}} \leq X$. Take expectations and divide by the positive number $a$. □

**A picture proof: fit a rectangle under the tail.** For $Y \geq 0$, the tail identity says that $\mathbb{E}\,Y$ is the area under the nonincreasing curve $S(t) = P(Y > t)$:

$$\mathbb{E}\,Y = \int_0^\infty S(t)\,dt.$$

Choose a threshold $a > 0$. Draw the rectangle with opposite corners $(0,0)$ and $(a, S(a))$. As $a$ changes, why does this rectangle always stay below the curve?

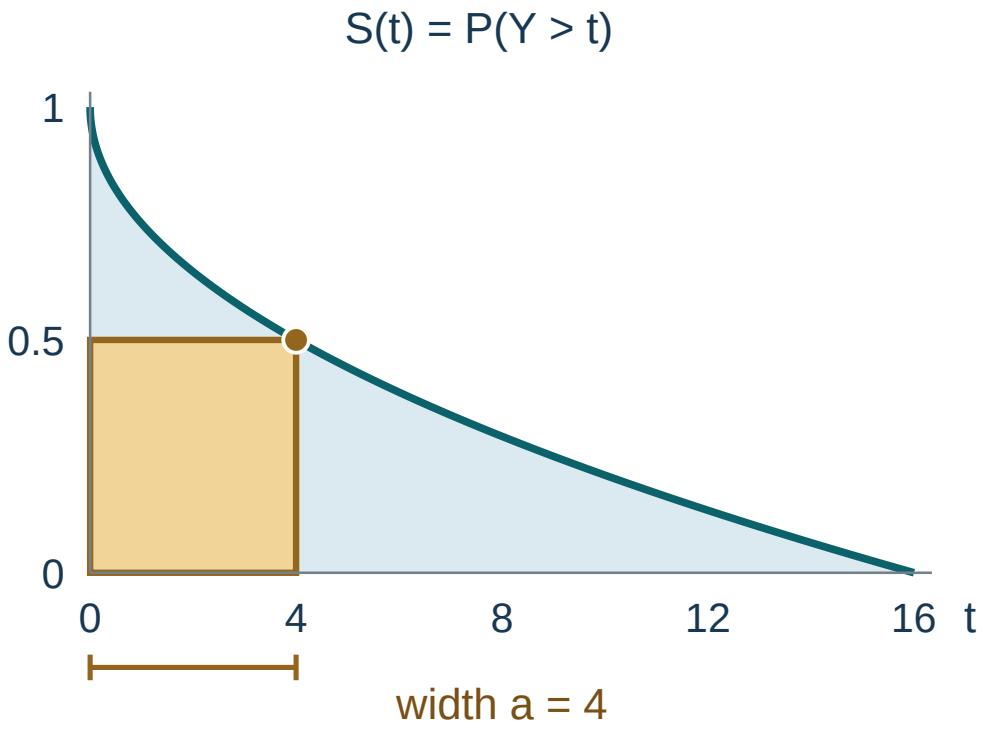
For the picture, take $Y = X^2$ with $X \sim \mathrm{Unif}([0,4])$. For $0 \leq t \leq 16$,

$$S(t) = P(X^2 > t) = P(X > \sqrt{t}) = 1 - \frac{\sqrt{t}}{4},$$

and $S(t) = 0$ for $t > 16$. The whole shaded area is

$$\mathbb{E}\,Y = \mathbb{E}\,X^2 = \frac{1}{4}\int_0^4 x^2\,dx = \frac{16}{3}.$$

At $a = 4$, the rectangle has width 4, height $P(Y > 4) = 1/2$, and area $2 \leq 16/3$.




The geometric reason works for every nonnegative $Y$: for $0 \le t < a$, $S(t) \ge S(a)$. Comparing areas gives

$$\underbrace{\mathbb{E}\,Y}_{\text{whole area}} = \int_0^\infty S(t)\,dt \ge \int_0^a S(t)\,dt \ge \underbrace{aP(Y > a)}_{\text{rectangle area}}.$$

Dividing by $a$ yields $P(Y > a) \le \mathbb{E}\,Y/a$. For the theorem's $P(Y \ge a)$ version, use $\{Y \ge a\} \subseteq \{Y > t\}$ for every $t < a$: the rectangle of height $P(Y \ge a)$ also fits under the curve on $[0, a)$, so $\mathbb{E}\,Y \ge aP(Y \ge a)$. A single endpoint does not change its area.

Applying Markov to the nonnegative variable $(X - \mathbb{E}\,X)^2$ gives **Chebyshev's inequality**, for $X \in L^2(P)$ and $t > 0$:

$$P(|X - \mathbb{E}\,X| \ge t) \le \frac{\mathrm{Var}(X)}{t^2}.$$

This is a probability bound from a moment, whereas the tail identity recovers a moment from all tail probabilities. A bound greater than one can always be replaced by one; it is valid but uninformative.

Used again in weak laws, Chernoff bounds.

## 2.3 Almost sure convergence and expectations

**High priority** *Keep the exceptional set and the envelope explicit*

We say $X_n \to X$ **almost surely (a.s.)** if there is a measurable $N$ with $P(N) = 0$ such that $X_n(\omega) \to X(\omega)$ for every $\omega \notin N$. This is pointwise convergence outside a single null set, not necessarily convergence at every outcome. We write $X_n \to X$ a.s., rather than suppressing the mode of convergence.

**Instructor's Note** *Follow one outcome at a time*

Think of each outcome $\omega$ as choosing a path $X_1(\omega), X_2(\omega), \ldots$. Almost sure convergence says that the path settles toward $X(\omega)$ for almost every choice. We can set the values on the exceptional null set to zero and picture convergence along every path. To understand convergence of averages, also picture the integrable envelope $Y$ in the DCT below: it keeps the paths from making ever taller spikes in ever smaller regions, as in Lecture 3.

For the dominated convergence theorem, suppose $X_n \to X$ a.s. and $|X_n| \le Y$ a.s. for every $n$, with $Y \ge 0$ and $\mathbb{E}\, Y < \infty$. Recall the definition of integrability. Outside the countable union of the exceptional sets, $|X| \le Y$ and $|X_n - X| \le 2Y$. The DCT from Lecture 3 therefore gives

$$\mathbb{E}\,|X_n - X| \longrightarrow 0, \qquad |\,\mathbb{E}\, X_n - \mathbb{E}\, X| \le \mathbb{E}\,|X_n - X| \longrightarrow 0.$$

The first assertion is convergence in the $L^1$ norm defined above.

# 3 Useful examples

## 3.1 The moment method for random graphs

**Low priority** *Count structures with indicators*

In the **Erdős–Rényi random graph** $G \sim \mathcal{G}(n, p_n)$, the vertex set is $\{1, \ldots, n\}$ and the indicators $A_{ij}$ for $1 \le i < j \le n$ are independent Bernoulli variables with success probability $p_n$. Set $A_{ji} = A_{ij}$ and $A_{ii} = 0$: opposite orientations of one undirected edge are the same variable, not independent copies. Let $B_{ijk}$ be the event that vertices $i < j < k$ form a triangle. Then

$$N = \sum_{i<j<k} \mathbf{1}_{B_{ijk}}, \qquad \mathbb{E}\, N = \sum_{i<j<k} P(B_{ijk}) = \binom{n}{3} p_n^3.$$

Linearity needs no independence among the triangle indicators. Independence of the three distinct edges gives $P(B_{ijk}) = p_n^3$. Markov yields

$$P(N \ge 1) \le \min\{1, \mathbb{E}\, N\}.$$

For $p_n = n^{-\alpha}$ with fixed $\alpha > 0$, $\mathbb{E}\, N = (1 + o(1))n^{3-3\alpha}/6$. Hence for $\alpha > 1$ the probability of no triangle tends to one. A diverging expected count when $\alpha < 1$ *does not by itself* show that a triangle is likely: variability must also be controlled. A second-moment calculation in Exercise 9 and its solution supplies the missing argument and gives

$$P(N = 0) = \begin{cases} 1 - o(1), & \alpha > 1, \\ o(1), & 0 < \alpha < 1. \end{cases}$$

The boundary $\alpha = 1$ has $\mathbb{E}\, N \to 1/6$; these two off-boundary arguments do not determine its limiting probability of no triangle.

## 3.2 KL divergence, entropy, and cross-entropy

**Mid priority** *Compare probability laws through a density ratio*

Let $P, Q$ be probability measures on $(\mathbb{R}^d, \mathcal{B}^d)$ with $P \ll Q$ and let $r = dP/dQ$ be their Radon–Nikodym derivative. Recall that this means $P(B) = \int_B r\, dQ$, with $r \ge 0$ and $\int r\, dQ = 1$. The **Kullback–Leibler (KL) divergence** is

$$D_{\mathrm{KL}}(P\|Q) = \int \log r\, dP = \int r \log r\, dQ,$$

where $0 \log 0 = 0$. The value of $\log r$ on $\{r = 0\}$ is irrelevant under $P$. This integral is well defined, possibly $+\infty$, because $(r \log r)^- \le 1/e$ and $Q$ is a probability measure. If $P \not\ll Q$, we set $D_{\mathrm{KL}}(P\|Q) = +\infty$. For Lebesgue densities $f, g$ it becomes

$$D_{\mathrm{KL}}(P\|Q) = \int f(x) \log \frac{f(x)}{g(x)}\, dx,$$

with a zero contribution where $f = 0$ and an infinite contribution if $g = 0$ on a set of positive $P$-probability.

The convex-function idea in Jensen's inequality applies to $h(u) = u \log u$, extended by $h(0) = 0$ on $[0, \infty)$. To avoid a hidden domain or finite-integral assumption, use its explicit supporting-line gap:

$$u \log u - u + 1 \ge 0 \quad (u \ge 0), \qquad u \log u - u + 1 = 0 \iff u = 1.$$

For $u > 0$ its derivative is $\log u$ and its second derivative is $1/u$; at $u = 0$ the gap is one. Consequently,

$$D_{\mathrm{KL}}(P\|Q) = \int (r \log r - r + 1)\, dQ \ge 0.$$

Equality holds exactly when $r = 1$ $Q$-a.e., which is equivalent to $P = Q$ as measures. "$P = Q$ almost surely" is not the appropriate statement. KL is generally asymmetric, so it is not a metric, although it is a useful measure of discrepancy.

If $P$ has pmf $p_1, \dots, p_K$ and $Q$ is uniform on $\{1, \dots, K\}$,

$$D_{\mathrm{KL}}(P\|Q) = \sum_{k:p_k>0} p_k \log(K p_k) = \log K - H(P), \qquad H(P) = -\sum_{k=1}^{K} p_k \log p_k.$$

Here $H(P)$ is the **entropy**. In particular, $H(P) \le \log K$, with equality exactly for the uniform law. For a Bernoulli law with parameter $\gamma$, the Lecture 1 formula is

$$H(P) = -\gamma \log \gamma - (1 - \gamma) \log(1 - \gamma),$$

including the endpoints by $0 \log 0 = 0$. All logarithms here are natural.

For target probabilities $p_k$ and predicted probabilities $q_k$ over $K$ classes, the **cross-entropy** loss is

$$H(p, q) = \mathbb{E}_{Y \sim p}[-\log q_Y] = -\sum_{k:p_k>0} p_k \log q_k = H(p) + D_{\mathrm{KL}}(p\|q).$$

If $q_k = 0$ while $p_k > 0$, both the loss and divergence are infinite. Since $H(p)$ is fixed during prediction, minimizing cross-entropy is the same as minimizing this KL divergence. Real-valued scores $z_1, \dots, z_K$, often called logits, produce positive predicted probabilities through the **softmax function**

$$q_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad k = 1, \dots, K.$$

**Exercise prompt: binary softmax and logistic loss.** For classes $0, 1$, set $z_0 = 0, z_1 = z$ and let the observed label be $y \in \{0, 1\}$. Derive $q_1$ and simplify the cross-entropy for this single label to the usual logistic loss. See Exercise 5 in Section 5 and its complete solution.

Used again in cross-entropy and likelihood, likelihood consistency.

### 3.3 Collinearity and covariance geometry

**Low priority** *Center before interpreting a zero-variance direction*

In regression, regard $X = (X_1, \dots, X_p)^\top$ as a random feature vector and observations as its realized values. A nontrivial deterministic relation $a^\top X = 0$ a.s. puts its law on a linear subspace. More generally, $a^\top X = c$ a.s. puts the law on an affine hyperplane. Duplicate features can cause the first relation; a full set of category indicators sums to one and gives the second.

Let $m = \mathbb{E}\, X$ and $\Sigma = \mathbb{E}[(X-m)(X-m)^\top]$, assuming finite second moments. If $\operatorname{rank}(\Sigma) = r < p$, write $\Sigma = U\Lambda U^\top$, where $U$ is $p \times r$, $U^\top U = I_r$, and $\Lambda$ contains the positive eigenvalues. For $U^\top u = 0$,

$$\operatorname{Var}(u^\top X) = u^\top \Sigma u = 0, \qquad u^\top (X - m) = 0 \quad \text{a.s.}$$

Using a finite orthonormal basis of the orthogonal complement shows

$$X = m + U\widetilde{X} \quad \text{a.s.}, \qquad \widetilde{X} = U^\top (X - m).$$

Thus the law is concentrated on $m + \operatorname{range}(U)$, an affine subspace. The complete argument and a translated-line example are in Exercise 10 and its solution. Centering locates the subspace; $U^\top X$ alone omits the reconstruction's mean term.

For observations $X^{(1)}, \dots, X^{(n)} \in \mathbb{R}^p$, let $\bar{X} = n^{-1} \sum_i X^{(i)}$ and let $\mathsf{X}_c$ be the $n \times p$ matrix with rows $(X^{(i)} - \bar{X})^\top$. The course's empirical covariance convention is

$$\widehat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^\top = \frac{\mathsf{X}_c^\top \mathsf{X}_c}{n}. \tag{5}$$

The factor $1/(n-1)$ is another convention, but is not used here. If the population mean is known to be zero and centering is not performed, $n^{-1} \sum_i X^{(i)} (X^{(i)})^\top$ estimates the second moment, which then equals the population covariance. Otherwise it estimates a different matrix. PCA finds directions of greatest empirical variance. Since $\operatorname{rank}(\widehat{\Sigma}) \le \min(p, n-1)$, sample rank deficiency alone does not establish population collinearity. A small-variance direction also need not be irrelevant for predicting a response.

### Key ideas

- Expectation can be computed on the original sample space or through the law of a random variable.
- Tail probabilities encode the expectation of a nonnegative variable.
- Jensen, Hölder, Markov and Chebyshev turn structural assumptions into useful inequalities.
- Limit exchanges and information inequalities depend on their stated integrability conditions.

### Looking ahead

Lecture 5 asks how averaging changes after we observe additional information. Expectation and the squared-error inequalities here lead to conditional means and optimal prediction. Continue to Lecture 5.

## 4 Further reading

**Further reading** *Measure decomposition and complete function spaces*

For two measures, **mutual singularity** $\nu \perp \eta$ means that some measurable set $S$ satisfies $\nu(S^c) = 0$ and $\eta(S) = 0$. These are disjoint sets of concentration, not necessarily disjoint *topological supports*. A set of concentration need not be unique or be the largest set on which the measure is positive.

Here is the construction behind (3). For each integer $k \geq 1$, there are at most $k$ atoms of mass at least $1/k$; hence their union $D$ is countable. Remove the atomic measure to form the nonnegative atomless measure $\nu = \mu - \mu_{\text{dis}}$. Apply the Lebesgue decomposition theorem to $\nu$ relative to Lebesgue measure $\lambda$: $\nu = \mu_{\text{cont}} + \mu_{\text{sing}}$, where $\mu_{\text{cont}} \ll \lambda$ and $\mu_{\text{sing}} \perp \lambda$. Both components are atomless because each is bounded by $\nu$. If $S$ is a Lebesgue-null set carrying $\mu_{\text{sing}}$, remove $D$ from it without changing its $\mu_{\text{sing}}$ mass. Then

$$\mu(A) = (\mu_{\text{dis}} + \mu_{\text{cont}})(A \cap S^c) + \mu_{\text{sing}}(A \cap S).$$

This recovers the original mutually exclusive-events picture precisely. The atoms are determined by $\mu$, and uniqueness of the two-part Lebesgue decomposition gives uniqueness of all three parts. A proof of the theorem used here is in John Hunter's *Measure Theory*, Theorem 6.27; see also Chapter 6 of [2]. The existence of the singular continuous component does not follow merely from the phrase "singular relative to the other two parts"; its atomless property is essential.

On almost-sure equivalence classes, $L^p(P)$ is a complete normed space, also called a **Banach space**,[3] for $1 \leq p \leq \infty$. The completeness proof is in Hunter's Theorem 7.10. Convergence in its norm is written

$$X_n \xrightarrow{L^p} X \quad \Longleftrightarrow \quad \|X_n - X\|_p \longrightarrow 0.$$

This is a different definition from almost sure convergence. Section 1.5.1 of [3] develops modes of convergence further. For KL divergence, entropy, and cross-entropy, Chapter 2 of [1] gives an information-theoretic perspective.

# 5 Exercises and complete solutions

**High priority** *Practice using a law*

**Exercise 1 (Change of variables through the law).** For a real random variable $X$ with law $\mu_X$, prove $\mathbb{E}\, g(X) = \int g \, d\mu_X$ first for a nonnegative simple Borel function $g = \sum_{j=1}^{m} c_j \mathbf{1}_{B_j}$ with disjoint $B_j$. Extend the identity to nonnegative Borel functions and then to integrable signed functions.

**Solution.** For an indicator, $\mathbb{E}\, \mathbf{1}_B(X) = P(X \in B) = \mu_X(B)$. Finite additivity of the integral now gives

$$\mathbb{E}\, g(X) = \sum_j c_j P(X \in B_j) = \sum_j c_j \mu_X(B_j) = \int g \, d\mu_X.$$

For a general nonnegative Borel $g$, take simple $g_n \uparrow g$. Then $g_n(X) \uparrow g(X)$, so applying MCT once under $P$ and once under $\mu_X$ gives

$$\mathbb{E}\, g(X) = \lim_n \mathbb{E}\, g_n(X) = \lim_n \int g_n \, d\mu_X = \int g \, d\mu_X.$$

This includes infinite values. Apply this identity to $|g|$ to see that integrability on the two spaces is equivalent. Finally apply it separately to $g^+$ and $g^-$ and subtract when both integrals are finite (or when at least one is finite for an extended expectation). The proof never uses injectivity or a density. □

**High priority** *Practice choosing an interchange theorem*

**Exercise 2 (Tail integrals, including infinite expectations).** Prove the main-text identity $\mathbb{E}\,|X| = \int_0^\infty P(|X| > t)\, dt$ without assuming finiteness. Derive (4) under its stated assumptions. As a numerical illustration, calculate both sides for a variable $Y$ equally likely to be 1 or 3.

[3] A Banach space is a normed vector space in which every Cauchy sequence converges in that norm to an element of the same space. "Cauchy" means that $\|X_n - X_m\|$ becomes arbitrarily small for all sufficiently large $n, m$.

**Solution.** For each fixed $y \geq 0$, $y = \int_0^\infty \mathbf{1}_{\{t<y\}}\, dt$. The function $(\omega, t) \mapsto \mathbf{1}_{\{t<|X(\omega)|\}}$ is measurable and nonnegative. Tonelli on $P \otimes dt$ therefore gives

$$\mathbb{E}\,|X| = \int_\Omega \int_0^\infty \mathbf{1}_{\{t<|X(\omega)|\}}\, dt\, dP = \int_0^\infty P(|X| > t)\, dt.$$

For the extension, the fundamental theorem of calculus gives $f(y) = \int_0^y f'(t)\, dt$. Since $f' \geq 0$, another application of Tonelli to $f'(t)\mathbf{1}_{\{t<|X(\omega)|\}}$ yields (4). Neither argument needs an a priori finite expectation. The value of the indicator at $t = y$ does not affect the Lebesgue integral in $t$. For the two-point example, $\mathbb{E}\,Y = (1+3)/2 = 2$, while

$$P(Y > t) = \begin{cases} 1, & 0 \leq t < 1, \\ 1/2, & 1 \leq t < 3, \\ 0, & t \geq 3. \end{cases} \qquad \int_0^\infty P(Y > t)\, dt = 1 \cdot 1 + \tfrac{1}{2} \cdot 2 = 2.$$

The rectangle picture is an illustration; Tonelli proves the identity for arbitrary distributions. □

**High priority** *Practice with moment orders*

**Exercise 3 (Higher moments imply lower moments).** Suppose $1 \leq p < q < \infty$ and $X \in L^q(P)$. Prove $\|X\|_p \leq \|X\|_q$. State the analogous conclusion for $q = \infty$.

**Solution.** Apply Hölder to $|X|^p \cdot 1$ with conjugate exponents $q/p$ and $q/(q-p)$. Its first factor is integrable to the required power because $(|X|^p)^{q/p} = |X|^q$. Since $P(\Omega) = 1$,

$$\mathbb{E}\,|X|^p \leq (\mathbb{E}\,|X|^q)^{p/q} (\mathbb{E}\, 1^{q/(q-p)})^{(q-p)/q} = (\mathbb{E}\,|X|^q)^{p/q}.$$

Taking the $p$th root gives the claimed bound and proves finiteness. If $X \in L^\infty$, for every $\varepsilon > 0$ we have $|X| \leq \|X\|_\infty + \varepsilon$ a.s.; hence $\|X\|_p \leq \|X\|_\infty + \varepsilon$. Let $\varepsilon \downarrow 0$. The constant one in these comparisons uses the fact that the underlying measure has total mass one. □

**High priority** *Practice with centered variables*

**Exercise 4 (Covariance bound and zero variance).** Let $X, Y \in L^2(P)$. Show that $|\mathrm{Cov}(X,Y)| \leq \sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$ and treat the case where one variance is zero.

**Solution.** Set $U = X - \mathbb{E}\,X$ and $V = Y - \mathbb{E}\,Y$. These are in $L^2$, and Cauchy–Schwarz gives

$$|\mathrm{Cov}(X,Y)| = |\,\mathbb{E}\,UV| \leq \mathbb{E}\,|UV| \leq (\mathbb{E}\,U^2)^{1/2} (\mathbb{E}\,V^2)^{1/2} = \sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}.$$

If $\mathrm{Var}(X) = \mathbb{E}\,U^2 = 0$, then $U = 0$ a.s.: for each $k \geq 1$, $P(|U| \geq 1/k) \leq k^2\, \mathbb{E}\,U^2 = 0$, and their union is $\{|U| > 0\}$. Thus $\mathrm{Cov}(X,Y) = 0$ for every such $Y$; the same applies when $\mathrm{Var}(Y) = 0$. If both variances are positive, dividing by their standard deviations shows that the correlation is in $[-1, 1]$. Correlation is not defined by that quotient when one variance is zero. □

**Mid priority** *Practice interpreting a classification loss*

**Exercise 5 (Binary softmax gives logistic loss).** Use scores $z_0 = 0, z_1 = z$ for classes $0, 1$. Derive the softmax probability $q_1$ and the cross-entropy for an observed class $y \in \{0, 1\}$. Also give the formula for a target probability $\gamma \in [0, 1]$ of class 1.

**Solution.** The two probabilities are $q_1 = e^z/(1+e^z) = 1/(1+e^{-z})$ and $q_0 = 1/(1+e^z)$. Hence $\log q_1 = z - \log(1+e^z)$ and $\log q_0 = -\log(1+e^z)$. Substituting into the loss gives

$$-y \log q_1 - (1-y) \log q_0 = \log(1+e^z) - yz.$$

For $\widetilde{y} = 2y - 1 \in \{-1, 1\}$ this equals $\log(1 + e^{-\widetilde{y}z})$, the binary logistic loss. A target probability $\gamma$ gives $\log(1 + e^z) - \gamma z$ by replacing $y$ with $\gamma$. More general scores yield the same probabilities after subtracting $z_0$ from both scores, so only their difference matters. □

**Mid priority** *Details for the Gaussian examples*

**Exercise 6 (One square and a squared radius).** Find the density of $Z^2$ for $Z \sim N(0, 1)$. For independent standard normals $Z_1, Z_2$, prove that $Z_1^2 + Z_2^2$ is exponential with rate $1/2$.

**Solution.** Writing $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$, for $t > 0$, $P(Z^2 \le t) = \int_{-\sqrt{t}}^{\sqrt{t}} \phi(z)\, dz$. Differentiating both endpoints gives

$$f_{Z^2}(t) = \frac{\phi(\sqrt{t}) + \phi(-\sqrt{t})}{2\sqrt{t}} = \frac{e^{-t/2}}{\sqrt{2\pi t}}, \qquad t > 0.$$

There is no atom at zero, since $P(Z = 0) = 0$; the density is zero for $t < 0$. This is the $\chi_1^2$ density. For the two-dimensional radius, independence gives the joint density $(2\pi)^{-1} e^{-(z_1^2 + z_2^2)/2}$. For $t \ge 0$, polar coordinates, including their Jacobian $r$, give

$$P(Z_1^2 + Z_2^2 > t) = \int_0^{2\pi} \int_{\sqrt{t}}^{\infty} \frac{1}{2\pi} e^{-r^2/2} r \, dr \, d\theta = e^{-t/2}.$$

Differentiating $1 - e^{-t/2}$ yields density $\frac{1}{2} e^{-t/2}$ on $(0, \infty)$. Its mean, from the tail identity, is $\int_0^\infty e^{-t/2}\, dt = 2$, consistent with $\mathbb{E}\, Z_1^2 + \mathbb{E}\, Z_2^2 = 2$. □

**High priority** *Details for the norm and inequality assertions*

**Exercise 7 (Hölder and the $L^p$ norm).** Prove Hölder's inequality and verify the norm properties of $L^p(P)$, $1 \le p \le \infty$, after identifying variables equal almost surely.

**Solution.** For $1 < p < \infty$ and $q = p/(p-1)$, Young's scalar inequality is $ab \le a^p/p + b^q/q$ for $a, b \ge 0$. To see it, for fixed $b$ minimize $a^p/p - ab$ over $a \ge 0$; its minimum occurs at $a = b^{1/(p-1)}$ and equals $-b^q/q$ (including $b = 0$). If $\|X\|_p, \|Y\|_q$ are positive, apply Young to $|X|/\|X\|_p$ and $|Y|/\|Y\|_q$ and integrate. The right side is $1/p + 1/q = 1$, which proves Hölder. If one norm is zero, that variable is zero a.s., so the result is immediate. For $p = 1, q = \infty$, the bound follows from $|Y| \le \|Y\|_\infty + \varepsilon$ a.s. and then $\varepsilon \downarrow 0$; the other endpoint is symmetric.

Nonnegativity and homogeneity follow directly from the definitions. For finite $p$, norm zero implies $X = 0$ a.s. because $P(|X| \ge 1/k) \le k^p\, \mathbb{E}\, |X|^p = 0$ for all $k$. For $p = \infty$, norm zero implies $P(|X| > 1/k) = 0$ for all $k$, with the same conclusion. Conversely an a.s. zero variable has norm zero. This also proves that the value of a norm is unchanged by null-set modifications.

It remains to prove the triangle inequality. For $p = 1$, integrate $|X + Y| \le |X| + |Y|$. For $p = \infty$, combine a.s. bounds with $\varepsilon > 0$ and let $\varepsilon \downarrow 0$. For $1 < p < \infty$, first use $|X + Y|^p \le 2^{p-1}(|X|^p + |Y|^p)$, a scalar convexity inequality, to establish $X + Y \in L^p$. Then Hölder gives

$$\begin{aligned} \mathbb{E}\, |X + Y|^p &\le \mathbb{E}\left[|X||X + Y|^{p-1}\right] + \mathbb{E}\left[|Y||X + Y|^{p-1}\right] \\ &\le (\|X\|_p + \|Y\|_p)\left(\mathbb{E}\, |X + Y|^{(p-1)q}\right)^{1/q} \\ &= (\|X\|_p + \|Y\|_p)\|X + Y\|_p^{p-1}. \end{aligned}$$

If $\|X + Y\|_p = 0$ the triangle inequality is immediate; otherwise divide by its finite positive $(p-1)$st power. This proves Minkowski without assuming the norm property one is trying to establish. Together with linearity of almost-sure equivalence classes, these facts give the claimed normed vector space. Completeness is a separate theorem, referenced in the further reading. □

**Mid priority** *Spatial log-gradients and boundary conditions*

**Exercise 8 (A justified zero mean for the spatial log-gradient).** Let $f \geq 0$ be a $C^1$ density on $\mathbb{R}^d$, zero outside a compact set. Define $S(x) = \nabla f(x)/f(x)$ if $f(x) > 0$ and $S(x) = 0$ otherwise. Prove that every component of $S(X)$ is integrable and $\mathbb{E}\, S(X) = 0$. Give one smooth-at-the-boundary one-dimensional example.

**Solution.** At a point where $f = 0$, nonnegativity makes that point a differentiable minimum, so $\nabla f = 0$. For each coordinate $j$,

$$\mathbb{E}\,|S_j(X)| = \int_{\{f>0\}} |\partial_j f(x)|\, dx = \int_{\mathbb{R}^d} |\partial_j f(x)|\, dx < \infty,$$

because the derivative is continuous and compactly supported. Thus

$$\mathbb{E}\, S_j(X) = \int_{\mathbb{R}^d} \partial_j f(x)\, dx.$$

Choose $R$ so that the support is inside $(-R, R)^d$. For fixed other coordinates, the fundamental theorem of calculus gives

$$\int_{-R}^{R} \partial_j f(x)\, dx_j = f(x_1, \ldots, R, \ldots, x_d) - f(x_1, \ldots, -R, \ldots, x_d) = 0.$$

Absolute integrability permits Fubini to integrate this zero over the other coordinates. Thus every component of $\mathbb{E}\, S(X)$ is zero. The calculation uses integration of a spatial derivative and vanishing boundary terms, not differentiation of a scalar normalizing constant with respect to $x$. For example, $f(x) = 30x^2(1-x)^2\mathbf{1}_{(0,1)}(x)$ integrates to one, since $\int_0^1 (x^2 - 2x^3 + x^4)\, dx = 1/30$. Both $f$ and its interior derivative vanish at the endpoints, so the zero extension is $C^1$. On $(0, 1)$ the log-derivative is $2/x - 2/(1-x)$. Although it is unbounded, it is integrable under this density by the argument above, and its expectation is $f(1) - f(0) = 0$. In contrast, for the density $f(x) = 2x\mathbf{1}_{(0,1)}(x)$, the zero extension jumps at 1, and $\mathbb{E}[(\log f)'(X)] = \int_0^1 (1/x)\, 2x\, dx = 2$. The nonzero boundary contribution $f(1-) - f(0+) = 2$ explains why the smooth-boundary condition matters. □

**Low priority** *Details for the optional random-graph threshold*

**Exercise 9 (The missing second-moment argument).** For $G \sim \mathcal{G}(n, p_n)$, let $N$ count triangles. Compute its variance and use it to show $P(N = 0) \to 0$ if $p_n = n^{-\alpha}$, $0 < \alpha < 1$.

**Solution.** Each triangle indicator has variance $p_n^3 - p_n^6$. Distinct triangles with no common edge depend on disjoint sets of independent edge variables and hence have covariance zero, even if they share a vertex. Those sharing one edge have five distinct required edges, so covariance $p_n^5 - p_n^6$. There are $\binom{n}{2}\binom{n-2}{2} = 6\binom{n}{4}$ unordered pairs sharing an edge. Since a variance expansion counts each unordered pair twice,

$$\mathrm{Var}(N) = \binom{n}{3}(p_n^3 - p_n^6) + 12\binom{n}{4}(p_n^5 - p_n^6).$$

For $n \geq 3$ and $p_n > 0$, Chebyshev, with $m_n = \mathbb{E}\, N > 0$, gives

$$P(N = 0) \leq P(|N - m_n| \geq m_n) \leq \frac{\mathrm{Var}(N)}{m_n^2} \leq \frac{1}{\binom{n}{3} p_n^3} + \frac{12\binom{n}{4}}{\binom{n}{3}^2 p_n}.$$

As $n \to \infty$, the terms are respectively $O(n^{-3+3\alpha})$ and $O(n^{-2+\alpha})$, both tending to zero for $0 < \alpha < 1$. This proves the claimed existence of a triangle with probability tending to one. The $\alpha > 1$ assertion follows from the first moment as in the main text; no critical-case probability was proved. □

**Low priority** *Details for covariance geometry*

**Exercise 10 (Affine concentration and centered empirical covariance).** Under the finite-second-moment and eigendecomposition assumptions of Section 3.3, prove $X = m + UU^\top(X - m)$ a.s. Check the empirical covariance formula and rank bound. Illustrate why "linear subspace" needs centering.

**Solution.** Choose an orthonormal basis $v_1, \dots, v_{p-r}$ for the orthogonal complement of the columns of $U$. For each $j$,

$$\mathbb{E}\left[(v_j^\top(X - m))^2\right] = v_j^\top \Sigma v_j = 0.$$

The zero-expectation argument from Exercise 4 shows that each scalar vanishes a.s. A finite union of the exceptional null sets is null, so all coordinates vanish simultaneously outside one null set. The orthogonal projection onto the complement is $I - UU^\top = \sum_j v_j v_j^\top$, giving $(I - UU^\top)(X - m) = 0$ a.s. If $r = 0$, the same argument gives $X = m$ a.s.

Matrix multiplication gives $\mathsf{X}_c^\top \mathsf{X}_c = \sum_i (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^\top$, so division by $n$ proves (5). Since $\mathbf{1}_n^\top \mathsf{X}_c = 0$, the columns lie in the $(n-1)$-dimensional orthogonal complement of $\mathbf{1}_n$. Also $\ker(\mathsf{X}_c^\top \mathsf{X}_c) = \ker(\mathsf{X}_c)$ because $v^\top \mathsf{X}_c^\top \mathsf{X}_c v = \|\mathsf{X}_c v\|^2$. Thus $\operatorname{rank}(\widehat{\Sigma}) = \operatorname{rank}(\mathsf{X}_c) \le \min(p, n-1)$. Let $T$ take $-1$ and $1$ equally likely and let $X = (T, T+2)^\top$. Then $m = (0, 2)^\top$ and

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Lambda = (2).$$

The two outcomes $(-1, 1)$ and $(1, 3)$ lie on the affine line $x_2 = x_1 + 2$, while $X - m$ lies on the linear line $x_2 = x_1$. For $u = (1, -1)^\top/\sqrt{2}$, $u^\top X = -\sqrt{2}$ is constant, not zero. Zero variance means constancy; centering turns that constant into zero. □

# References


[1] Thomas M. Cover and Joy A. Thomas. *Elements of information theory.* Wiley-Interscience, 2nd edition, 2006.

[2] W. Rudin. *Real and Complex Analysis.* Higher Mathematics Series. McGraw-Hill Education, 1987.

[3] Jun Shao. *Mathematical Statistics.* Springer Science & Business Media, 2nd edition, 2013.


**What changed**

**October 14, 2026.** Added issue-reporting links to the webpage and PDF. The mathematics and priority assignments are unchanged in this update.